Program : **Diploma in Automation and Robotics**	
Course Code : **5337**	Course Title: **Industrial Automation Control Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- To design electronic converters and electronic controllers using op-amp.
- To demonstrate the various characteristics of controllers
- To perform PLC programming using ladder diagram
- To simulate various controllers and step response using MATLAB & Simulink.

Course Prerequisites:

Topic	Course code	Course name	Semester
Computer knowledge		Introduction to IT System, Python programming	1,3
Operational Amplifiers		Linear IC and Power Electronics	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Implement error detector, Instrumentation Amplifier, Voltage to Current converter, Current to Voltage converter. Different controller modes (P, PI, PD, PID). et	9	Applying
CO2	Demonstrate the various characteristics of control valves	9	Applying
CO3	Practice preparation of ladder diagram and PLC Programming.	12	Applying

CO4	To simulate various controllers and step response using MATLAB & Simulink	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2	3						
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Implement analog Electronic Controllers and Converters**		
M1.01	Design and setup Error detector circuit andInstrumentation amplifier using Op-amp.	3	Applying
M1.02	Design and setup current to voltage and voltage tocurrent converters using OP amp.	3	Applying
M1.03	Design and setup P, PI, PD and PID controllersusing OP amp.	3	Applying
CO2	**Demonstrate the various characteristics of various process elements**		
M2.01	To Study the level characteristics when it is controlled by PID controller.	3	Applying
M2.02	To study the changing pattern of flow rate characteristics of process fluid	3	Applying
M2.03	To observe the Pressure Control characteristics.	3	Applying
	Lab Test – II	2	

CO3	**To perform PLC programming using ladder diagram**		
M3.01	Understand the concept of latch and de-latch in PLC programming and ladder diagram	3	Applying
M3.02	Upload Ladder logic to PLC memory debug and run the program Use ladder diagram for level control and position control	6	Applying
M3.03	Practice timer programming in PLC	3	Applying
CO4	**Simulate various controllers and step response using MATLAB & Simulink.**		
M4.01	Simulate Proportional and Proportional Integralcontroller using MATLAB & SIMULINK.	3	Applying
M4.02	Simulate Proportional derivative and ProportionalIntegral derivative controller using MATLAB &Simulink.	3	Applying
M4.03	Simulate instrumentation amplifier using MATLAB &Simulink.	3	Applying
M4.04	Flow control process/temperature control process: plot the response curve forstep response of set point change. Simulate for standard test signal	3	Applying
	Lab Test – II	1	

Text / Reference

T/R	Book Title/Author
T1	Industrial Instrumentation, Control and Automation, S. Mukhopadhyay, S. Sen and A. K. Deb, Jaico Publishing House, 2013
R1	Jonathan Love, Process Automation Handbook, Springer 2007
R1	Curtis. D. Johnson, Process control instrumentation technology, PHI.
R2	Liptak, Volume II, Instrument Engineers Hand book, Chilton Book Company.
R3	Krishnakanth, Computer based Industrial Control, PHI.
R4	Donald .P. Eckman, Automatic process control, WILLEY India.

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108105088
2	https://onlinecourses.nptel.ac.in/noc21_me67 /preview
3	https://scienceq.org/wp-content/uploads/2018/10/JAETV7I102.pdf
4	https://processcontrol.tv/tutorials
5	https://www.pidlab.com/en/virtual-labs